

Autonomous Multi-Agent Reddit Lead Generation System for Zeenie AI Employees

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1 Project Overview

This project implements an autonomous multi-agent lead generation pipeline that continuously discovers potential customers from Reddit, qualifies opportunities, recommends suitable Zeenie AI Employee solutions, generates personalized outreach messages, stores leads in a persistent database, and delivers executive reports automatically.

The objective is to replace manual prospecting with a scalable AI-driven workflow.

2 Problem Statement

Traditional lead generation requires:

- Monitoring multiple Reddit communities
- Reading large numbers of posts
- Identifying business pain points
- Scoring opportunities manually
- Writing personalized outreach messages
- Tracking discovered leads

This process is repetitive, time consuming, and difficult to scale.

The goal of this project is to automate the complete pipeline using specialized AI agents coordinated through a manager agent.

3 System Architecture

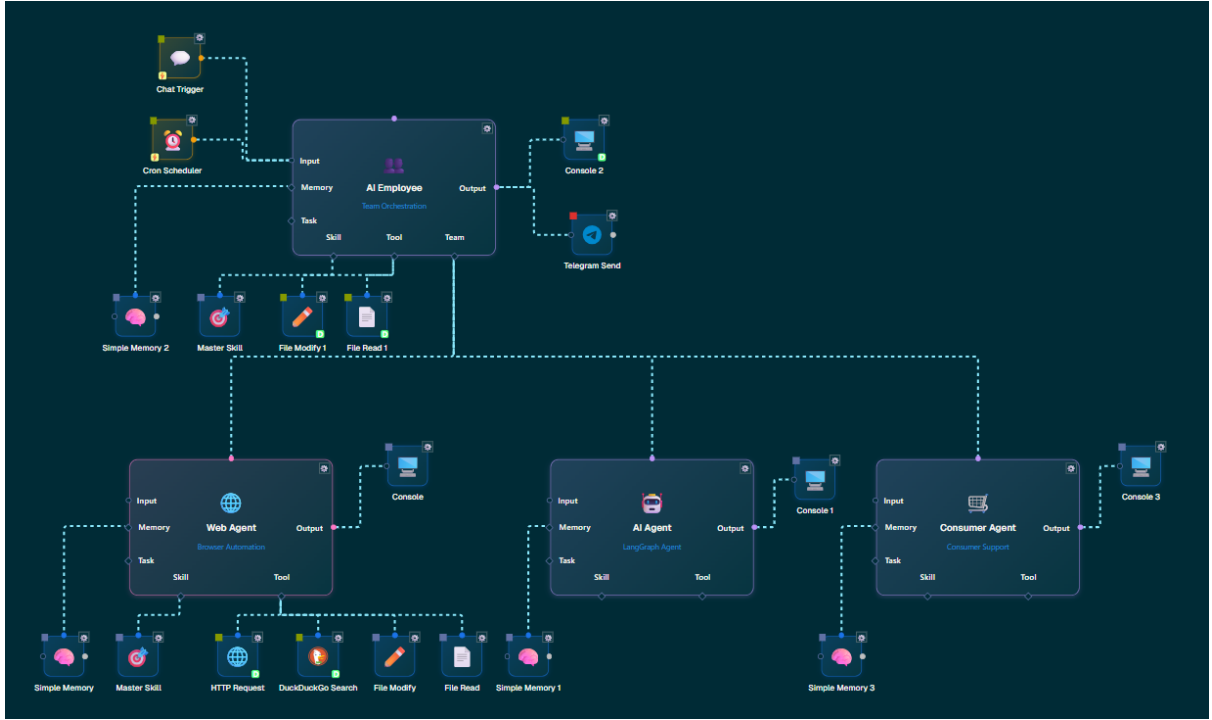


Figure 1: Complete Workflow Architecture

The system follows a hierarchical multi-agent architecture consisting of one manager agent and three specialist agents.

- Lead Generation Manager
- Reddit Hunter
- Lead Analyst
- Outreach Writer

Each agent is responsible for a specific task and operates independently.

4 Workflow Overview

The workflow executes as follows:

1. Trigger activated by user request or daily cron schedule.
2. Manager delegates Reddit discovery to Reddit Hunter.
3. Reddit Hunter collects recent opportunities.
4. Lead Analyst evaluates and scores opportunities.
5. Outreach Writer creates personalized messages.

6. Manager aggregates all outputs.
7. Leads are stored in JSON database.
8. Final report is sent to Telegram.

5 Node-by-Node Explanation

5.1 Chat Trigger

Purpose:

- Accepts manual user requests.
- Starts workflow execution.

Example:

Find founders looking for automation software.

5.2 Cron Scheduler

Purpose:

- Runs automatically every day at 9:00 AM.
- Enables autonomous lead monitoring.

Benefits:

- Fully automated prospect discovery.
- No human intervention required.

5.3 Lead Generation Manager (AI Employee)

Purpose:

- Acts as team leader.
- Coordinates specialist agents.
- Aggregates results.
- Generates executive report.
- Maintains lead database.

Responsibilities:

1. Delegate discovery.
2. Delegate analysis.

3. Delegate outreach generation.
4. Merge outputs.
5. Store leads.
6. Send final report.

5.4 Simple Memory

Purpose:

- Maintains conversation context.
- Stores intermediate workflow state.
- Allows agents to remember previous steps.

5.5 Master Skill

Purpose:

- Provides reusable instructions.
- Defines common workflow behaviour.

5.6 File Read Node

Purpose:

- Reads previously discovered leads.
- Prevents duplicate storage.
- Loads lead history from `data/leads.json`.

5.7 File Modify Node

Purpose:

- Stores newly discovered leads.
- Updates lead database.
- Maintains valid JSON structure.

Stored Fields:

- Date
- Title
- URL

- Subreddit
- Lead Score
- Pain Point
- Recommended Zeenie Solution

6 Reddit Hunter Agent

6.1 Purpose

The Reddit Hunter is responsible for discovering opportunities.
It does not score or analyze leads.

6.2 Connected Tools

- HTTP Request
- DuckDuckGo Search

6.3 Responsibilities

- Search Reddit communities.
- Extract recent posts.
- Identify automation-related discussions.
- Return structured candidate leads.

7 Lead Analyst Agent

7.1 Purpose

The Lead Analyst evaluates discovered opportunities and determines whether Zeenie can solve the prospect's problem.

7.2 Scoring Framework

Metric	Weight
Buying Intent	40
Pain Severity	25
Automation Fit	20
Engagement	10
Freshness	5

7.3 Output

For every lead:

- Pain Point
- Business Context
- Buying Intent
- Automation Fit
- Lead Score
- Recommended Zeenie Solution

8 Outreach Writer Agent

Purpose:

- Create personalized first-contact messages.
- Avoid generic sales language.
- Reference prospect's Reddit post.

Output:

- Personalized Opening Line
- Outreach Message
- CTA

9 Lead Storage Mechanism

All discovered opportunities are stored inside:

`data/leads.json`

Duplicate detection is performed using the Reddit URL as a unique key.
Workflow:

1. Read existing file.
2. Check URL.
3. Skip duplicates.
4. Append new lead.
5. Save updated file.

10 Telegram Notification System

After report generation:

1. Manager creates executive summary.
2. Report sent to Telegram.
3. Zeenie team receives daily opportunities.

11 Results

The system successfully:

- Discovers opportunities automatically.
- Qualifies prospects.
- Scores leads.
- Generates outreach.
- Stores leads.
- Runs on schedule.
- Sends notifications automatically.

12 Challenges Faced

- Agent recursion loops.
- Team coordination issues.
- Duplicate lead handling.
- Prompt engineering refinement.
- Reddit data quality.

13 Future Improvements

- Direct Reddit API integration.
- Multi-platform lead generation (LinkedIn, Twitter, Hacker News).
- CRM integration.
- Automatic outreach delivery.
- Lead conversion tracking.
- Better ranking models.

14 Conclusion

This project demonstrates how a multi-agent AI architecture can automate the complete lead generation lifecycle, from opportunity discovery to outreach generation.

The system reduces manual prospecting effort while improving scalability, consistency, and response speed for sales teams.